

Abstract

Cocoa (*Theobroma cacao*) is economically vital in West Africa but is severely threatened by the mirid bug *Sahlbergella singularis*, causing yield losses of 30–40% and eventual tree death. Despite its importance, the pest’s population dynamics and production impacts remain poorly quantified. This work synthesises three mathematical studies that build a comprehensive modelling framework. First, stage-structured ODE and delayed DDE models capture egg, nymph, and adult stages, with a basic offspring number determining extinction or persistence. Sensitivity analyses identify pod carrying capacity and adult mortality as key parameters; the delayed model matches seasonal rainfall patterns better, and combining mating disruption with trapping rivals chemical treatments with lower environmental cost. Second, a sex-structured piecewise-smooth DDE model with pheromone traps reveals two threshold quantities: below the lower, pheromones have no effect; above the upper, the pest is eradicated. This motivates a maximal-minimal strategy: intensive initial release drives the population to extinction, followed by minimal maintenance. Seasonal timing strongly affects efficiency. Third, coupling pest dynamics with a multi-stage pod model directly quantifies losses; simulations estimate 39% production loss without control. An optimal control problem, solved via Pontryagin’s minimum principle, shows that a 50-day intervention reduces losses to about 15%, yielding a financial gain of roughly 665 USD per plantation independent of initial pest density. Together, these studies demonstrate that rigorous mathematical modelling, calibrated with field data, provides actionable guidance for integrated pest management. Future work will incorporate spatial structure, time-varying delays, and multi-seasonal optimisation.

Keywords: *Sahlbergella singularis*; cocoa pest; stage-structured model; delay differential equations; piecewise-smooth system; sex-pheromone traps; optimal control; production losses; West Africa.